BRAKES

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Abbreviations

AAS	Active Adaptive Shift
ABDC	After Bottom Dead Center
ABS	Antilock Brake System
ACC	Accessories
AFS	Adaptive Front lighting System
ALC	Auto Level Control
ALR	Automatic Locking Retractor
ATDC	After Top Dead Center
ATF	Automatic Transaxle Fluid
ATX	Automatic Transaxle
BBDC	Before Bottom Dead Center
BCM	Body Control Module
BDC	Bottom Dead Center
BTDC	Before Top Dead Center
CAN	Controller Area Network
CCM	Comprehensive Component Monitor
CKP	Crankshaft Position
CM	Control Module
CMDTC	Continuous Memory Diagnostic Trouble Code
CMP	Camshaft Position
CMU	Connectivity Master Unit
CPU	Central Processing Unit
DAB	Digital Audio Broadcasting
DC	Drive Cycle
DEF	Defroster
DRSS	Distance Recognition Support System
DSC	Dynamic Stability Control
EBD	Electronic Brakeforce Distribution
EEPROM	Electrically Erasable Programmable Read-Only Memory
EHPAS	Electro Hydraulic Power Assist Steering
ELR	Emergency Locking Retractor
EPS	Electric Power Steering
ESS	Emergency Stop signal System
ESU	Electrical Supply Unit
EX	Exhaust
FBCM	Front Body Control Module
FSC	Forward Sensing Camera
GPS	Global Positioning System
HBC	High Beam Control
HF/TEL	Hands-Free Telephone
HI	High
HLA	Hydraulic Lash Adjuster
HLA	Hill Launch Assist
HP	High Pressure
HS	High Speed
HU	Hydraulic Unit
IDEVA	Intake stroke EGR using Double Exhaust Valve Actuation system
IDS	Integrated Diagnostic Software
IG	Ignition
IN	Intake
INT	Intermittent
KOEO	Key On Engine Off
KOER	Key On Engine Running
L.H.D.	Left Hand Drive
LCD	Liquid Crystal Display
LDWS	Lane Departure Warning System
LED	Light Emitting Diode
LF	Left Front

LH	Left Hand
LO	Low
LP	Low Pressure
LR	Left Rear
M	Motor
MAX	Maximum
MIN	Minimum
MRCC	Mazda Radar Cruise Control
MS	Middle speed
MTX	Manual Transaxle
NVH	Noise, Vibration, Harshness
OCV	Oil Control Valve
ODDTC	On-demand Diagnostic Trouble Code
P/W CM	Power Window Control Module
PAD	Passenger Air Bag Deactivation
PCV	Positive Crankcase Ventilation
PDS	Portable Diagnostic Software
PID	Parameter Identification
POWER MOS FET	Power Metal Oxide Semiconductor Field Effect Transistor
PSD	Power Sliding Door
PTC	Positive Temperature Coefficient
R.H.D.	Right Hand Drive
RBCM	Rear Body Control Module
RCTA	Rear Crossing Traffic Alert
RDP	Run Dry Prevention
RDS	Radio Data System
REC	Recirculate
RES	Rear Entertainment System
RF	Right Front
RH	Right Hand
RR	Right Rear
SAS	Sophisticated Air Bag Sensor
SCBS	Smart City Brake Support
SCR	Secondary Collision Reduction
SST	Special Service Tool
SW	Switch
TAU	Tuner and Amp Unit
TCS	Traction Control System
TDC	Top Dead Center
TFT	Transaxle Fluid Temperature
TNS	Tail Number Side Lights
TPMS	Tire Pressure Monitoring System
USB	Universal Serial Bus
VBC	Variable Boost Control
VENT	Ventilation
W/M	Workshop Manual
WGN	Wagon
1GR	First Gear
2GR	Second Gear
2WD	2-Wheel Drive
3GR	Third Gear
4GR	Fourth Gear
4SD	4 Door Sedan
4WD	4-Wheel Drive
5GR	Fifth Gear
5HB	5 Door Hatchback
6GR	Sixth Gear

Features

- A DSC HU/CM has been adopted in which the hydraulic unit (HU) and control module (CM) are integrated with a built-in brake fluid pressure sensor.

- Magnetic encoder type ABS sensor rotors have been adopted.
- Semiconductor element type ABS wheel-speed sensors have been adopted.
- The following systems, mechanisms, and functions have been adopted to enhance safety.
  - — Intrusion-minimizing brake pedal
  - — Antilock brake system (ABS)
  - — Electronic brakeforce distribution (EBD) control
  - — Traction control system (TCS)
  - — Dynamic stability control (DSC)
  - — Electronic control brake assist
  - — Vehicle roll prevention function
  - — Hill launch assist (HLA)
  - — Smart city brake support (SCBS) (With smart city brake support (SCBS))
- A vacuum pump has been adopted to compensate for the decrease in intake manifold vacuum applied to the power brake unit caused by the retard intake valve closing.
- Center lever type parking brake, adjustable from vehicle interior, has been adopted.

Specifications

Item		Specification
Brake pedal	Type	Suspended design
	Pedal lever ratio	2.75
Master cylinder	Type	Tandem
	Cylinder bore (mm {in})	22.22 {0.8748}
Front brake (disc)	Type	Ventilated disc
	Cylinder bore (mm {in})	53.96 {2.124}
	Pad dimensions (area x thickness) (mm^2 x mm {in^2 x in})	4,000 × 10 {8.200 × 0.39}
	Disc plate dimensions (mm {in})	258 × 22 {10.2 × 0.87}
Rear brake (disc)	Type	Solid disc
	Cylinder bore (mm {in})	34 {1.3}
	Pad dimensions (area x thickness) (mm^2 x mm {in^2 x in})	2,650 × 8.5 {4.108 × 0.33}
	Disc plate dimensions (mm {in})	255 × 9.5 {10.0 × 0.37}
Rear brake (drum)	Type	Leading/trailing type
	Wheel cylinder inner diameter (mm {in})	Except European (L.H.D. U.K.) specs. : 17.5 {0.689} European (L.H.D. U.K.) specs. : 19.1 {0.752}
	Lining dimensions (Length x Width x Thickness) (mm x mm x mm {in x in x in})	191.99 × 30 × 4.1 {7.5587 × 1.2 × 0.16}
	Drum inner diameter (mm {in})	200 {7.87}
Power brake unit	Type	Vacuum multiplier, single diaphragm
	Outer diameter (mm {in})	248.5 {9.783}
Rear wheel braking force control device	Type	Electronic Brakeforce Distribution (EBD)
Brake piping	Piping layout	X pattern
Parking brake	Type	Mechanical design, rear two-wheel braking
	Operating method (application/release)	Manually operated lever design
	Play adjustment method	Auto-adjusting
Brake fluid	Type	Rear brake (drum) [SKYACTIV-G 1.3 or SKYACTIV-G 1.5]: SAE J1703 or FMVSS116 DOT-3 or DOT-4 Rear brake (drum) [SKYACTIV-D 1.5]: SAE J1703 or FMVSS116 DOT-4 Rear brake (disc): SAE J1703 or FMVSS116 DOT-3 or DOT-4